

Ex. No: 1

Store and Display Student Names

PROGRAM:

```
import java.util.ArrayList;
import java.util.Scanner;

public class DisplayStudentNames {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        sc.nextLine();

        ArrayList<String> students = new ArrayList<String>();

        for (int i = 0; i < n; i++) {

            students.add(sc.nextLine());

        }

        for (String name : students) {

            System.out.println(name);

        }

        sc.close();

    }

}
```

Ex. No: 2

Find Total Number of Elements

PROGRAM:

```
import java.util.ArrayList;
import java.util.Scanner;

public class TotalNumberOfElements {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.nextInt());

        }

        System.out.println(list.size());

        sc.close();

    }

}
```

Ex. No: 3**Access Element Using Index****PROGRAM:**

```
import java.util.ArrayList;
import java.util.Scanner;

public class AccessElementIndex {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<String> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.next());

        }

        int index = sc.nextInt();

        System.out.println(list.get(index));

        sc.close();

    }

}
```

Ex. No: 4**Update an Element****PROGRAM:**

```
import java.util.*;

public class UpdateElement {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<String> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.next());

        }

        int index = sc.nextInt();

        String city = sc.next();

        list.set(index, city);

        for (String s : list) {

            System.out.println(s);

        }

    }

}
```

Ex. No: 5

Remove an Element by Value

PROGRAM:

```
import java.util.*;

public class RemoveElementByValue {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<String> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.next());

        }

        String fruit = sc.next();

        list.remove(fruit);

        for (String s : list) {

            System.out.println(s);

        }

    }

}
```

Ex. No: 6

Remove an Element by Index

PROGRAM:

```
import java.util.*;

public class RemoveElementByIndex {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.nextInt());

        }

        int index = sc.nextInt();

        list.remove(index);

        for (int x : list) {

            System.out.println(x);

        }

    }

}
```

Ex. No: 7

Check Element Availability

PROGRAM:

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<String> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.next());

        }

        String lang = sc.next();

        if (list.contains(lang))

            System.out.println("Available");

        else

            System.out.println("Not Available");

    }

}
```

Ex. No: 8

Print First and Last Element

PROGRAM:

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.nextInt());

        }

        System.out.println("First Element : " + list.get(0));

        System.out.println("Last Element : " + list.get(list.size() - 1));

    }

}
```

Ex. No: 9

Reverse Print the ArrayList

PROGRAM:

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.nextInt());

        }

        for (int i = list.size() - 1; i >= 0; i--) {

            System.out.println(list.get(i));

        }

    }

}
```

Ex. No: 10**Count Even Numbers****PROGRAM:**

```
import java.util.*;

public class CountEvenNumbers {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        int count = 0;

        for (int i = 0; i < n; i++) {

            int x = sc.nextInt();

            list.add(x);

            if (x % 2 == 0)

                count++;

        }

        System.out.println(count);

    }

}
```

Ex. No: 11**Find Maximum Element****PROGRAM:**

```
import java.util.*;

public class MaximumElement {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.nextInt());

        }

        int max = list.get(0);

        for (int x : list) {

            if (x > max)

                max = x;

        }

        System.out.println(max);

    }

}
```

Ex. No: 12**Find Minimum Element****PROGRAM:**

```
import java.util.*;

public class MinimumElement {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.nextInt());

        }

        int min = list.get(0);

        for (int x : list) {

            if (x < min)

                min = x;

        }

        System.out.println(min);

    }

}
```

Ex. No: 13**Calculate Sum of Elements****PROGRAM:**

```
import java.util.*;

public class SumOfElements {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        int sum = 0;

        for (int i = 0; i < n; i++) {

            int x = sc.nextInt();

            list.add(x);

            sum += x;

        }

        System.out.println(sum);

    }

}
```

Ex. No: 14**Search an Element****PROGRAM:**

```
import java.util.*;

public class SearchElement {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.nextInt());

        }

        int key = sc.nextInt();

        if (list.contains(key))

            System.out.println("Found");

        else

            System.out.println("Not Found");

    }

}
```

Ex. No: 15**Print Elements Greater Than a Given Number****PROGRAM:**

```
import java.util.*;

public class ElementsGreater {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.nextInt());

        }

        int k = sc.nextInt();

        for (int x : list) {

            if (x > k)

                System.out.println(x);

        }

    }

}
```

Ex. No: 16**Count Positive Numbers****PROGRAM:**

```
import java.util.*;

public class PositiveNumbers {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        int count = 0;

        for (int i = 0; i < n; i++) {

            int x = sc.nextInt();

            list.add(x);

            if (x > 0)

                count++;

        }

        System.out.println(count);

    }

}
```

Ex. No: 17**Print Elements at Even Indices****PROGRAM:**

```
import java.util.*;

public class ElementsEvenIndices {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list.add(sc.nextInt());

        }

        for (int i = 0; i < list.size(); i += 2) {

            System.out.println(list.get(i));

        }

    }

}
```

Ex. No: 18**Copy One ArrayList to Another****PROGRAM:**

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<Integer> list1 = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            list1.add(sc.nextInt());

        }

        ArrayList<Integer> list2 = new ArrayList<>(list1);

        for (int x : list2) {

            System.out.println(x);

        }

    }

}
```


